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AI Probability

1%

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LAPORAN PROYEK SIMULASI DAN PEMODELAN

Simulasi Ketidakpastian Jumlah Penumpang Kereta Api Jabodetabek Menggunakan Metode Monte Carlo berdasarkan data BPS tahun 2023-2026

Deksripsi Projek

Di wilayah jabodetabek, kereta api adalah salah satu sarana transportasi publik yang menjadi andalan utama masyarakat untuk bermobilitas harian.
Jumlah penumpang kereta api di Jabodetabek memiliki perubahan setiap bulannya.
Hingga perubahan untuk periode selanjutnya tidak bisa di prediksi secara pasti.
Di kondisi inilah menjadi dasar pemilihan studi kasus dalam proyek simulasi dan pemodelan.

Dengan mengambil basis data yang mengeksplorasi data historis penumpang kereta Jabodetabek rilisan Badan Pusat Statistik (BPS) dari Januari 2023 hingga Agustus 2026.
Data tersebut digunakan untuk melihat pola perubahan penumpang dan menjadi dasar dalam membuat simulasi penelitian menggunakan metode Monte Carlo.

Tujuan dalam simulasi ini sebagai perkiraan kemungkinan jumlah penumpang dan membandingkan hasil dengan data aktual, sehingga hasil yang diharapkan bisa menggambarkan ketidakpastian jumlah penumpang berdasarkan catatan periode sebelumnya.

Sumber Data

Dataset yang diekstrak dari table Jumlah Penumpang Kereta Api di laman resmi BPS (Tabel Dinamis - Badan Pusat Statistik Indonesia) ini mencakup periode pengamatan Januari 2023 sampai Agustus 2026.

Untuk pemrosesan awal, data diunduh dalam ekstensi CSV dengan satuan Kuantitas per ribu orang.

Dalam proyek ini data dibatasi dalam kategori wilayah Jabodetabek agar penumpang yang di analisis tetap konsisten.

Tahun

Periode

Jumlah Observasi

2023

Januari – Desember

12

2024

Januari – Desember

12

2025

Januari – Desember

12

2026

Januari – Agustus

8

Total

44

Metode Simulasi dan Pemodelan

Tahapan simulasi dilakukan beberapa tahap :

Mengolah data jumlah penumpang tahun 2023-2026 dari BPS.

Menghitung pola perubahan dari data historis setiap periode.

Mengambil sampel perubahan historis secara acak dengan pengulangan.

Melakukan 1.000 percobaan simulasi Monte Carlo.

Menghitung rata-rata hasil simulasi dan interval.

Hasil simulasi akan dilakukan perbandingan dengan data penumpang secara data aktual BPS.

Hasil Simulasi

Berikut hasil simulasi Monte Carlo dibandingkan dengan data aktual BPS menggunakan indikator MAE dan WAPE.

Indikator

Validasi 2025

Pengujian 2026

MAE

1390.21

1026.91

WAPE

4.78

3.41

Cakupan interval 80%

41.67

75.00

Hasil pengujian tahun 2026 menunjukkan WAPE sebesar 3.41%.
Artinya, total kesalahan absolut perkiraan setara dengan 3.41% dari jumlah penumpang aktual secara periode pengujian.

Hasil Simulasi Monte Carlo Penumpang Kereta Api Tahun 2026

Kesimpulan

Dari hasil simulasi Monte Carlo menggunakan data BPS 2023 – 2026 menunjukkan bahwa jumlah penumpang kereta api Jabodetabek mengalami perubahan setiap bulan.
Berdasarkan pengujian Januari – Agustus 2026, diperoleh WAPE sebesar 3.41%.
Hasil tersebut menunjukkan bahwa Metode Monte Carlo dapat digunakan untuk menggambarkan ketidakpastian jumlah penumpang, meskipun hasil perkiraannya masih memiliki selisih dengan data aktual.

High Human Impact High AI Impact

FAQs

What is GPTZero?

GPTZero is the leading AI detector for checking whether a document was written by a large language model such as ChatGPT. GPTZero detects AI on sentence, paragraph, and document level. Our model was trained on a large, diverse corpus of human-written and AI-generated text with support for English, Spanish, French, German, and other languages. To date, GPTZero has served over 20 million users around the world, and works with over 100 organizations in education, hiring, publishing, legal, and more.

When should I use GPTZero?

Our users have seen the use of AI-generated text proliferate into education, certification, hiring and recruitment, social writing platforms, disinformation, and beyond. We've created GPTZero as a tool to highlight the possible use of AI in writing text. In particular, we focus on classifying AI use in prose. Overall, our classifier is intended to be used to flag situations in which a conversation can be started (for example, between educators and students) to drive further inquiry and spread awareness of the risks of using AI in written work.

Does GPTZero only detect ChatGPT outputs?

No, GPTZero works robustly across a range of AI language models, including but not limited to ChatGPT, GPT-5, GPT-4, GPT-3, Gemini, Claude, and AI services based on those models.

What are the limitations of the classifier?

The nature of AI-generated content is changing constantly. As such, these results should not be used to punish students. We recommend educators to use our behind-the-scene Writing Reports as part of a holistic assessment of student work. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Instead, we recommend educators take approaches that give students the opportunity to demonstrate their understanding in a controlled environment and craft assignments that cannot be solved with AI. Our classifier is not trained to identify AI-generated text after it has been heavily modified after generation (although we estimate this is a minority of the uses for AI-generation at the moment). Currently, our classifier can sometimes flag other machine-generated or highly procedural text as AI-generated, and as such, should be used on more descriptive portions of text.

I'm an educator who has found AI-generated text by my students. What do I do?

Firstly, at GPTZero, we don't believe that any AI detector is perfect. There always exist edge cases with both instances where AI is classified as human, and human is classified as AI. Nonetheless, we recommend that educators can do the following when they get a positive detection: Ask students to demonstrate their understanding in a controlled environment, whether that is through an in-person assessment, or through an editor that can track their edit history (for instance, using our Writing Reports through Google Docs). Check out our list of several recommendations on types of assignments that are difficult to solve with AI.

Ask the student if they can produce artifacts of their writing process, whether it is drafts, revision histories, or brainstorming notes. For example, if the editor they used to write the text has an edit history (such as Google Docs), and it was typed out with several edits over a reasonable period of time, it is likely the student work is authentic. You can use GPTZero's Writing Reports to replay the student's writing process, and view signals that indicate the authenticity of the work.

See if there is a history of AI-generated text in the student's work. We recommend looking for a long-term pattern of AI use, as opposed to a single instance, in order to determine whether the student is using AI.